

HW 6

1

Determine the multiple-output minimal sums for each of the following sets of functions:

a

$$f_1 = yz + wz$$

$$f_2 = yz + wz + wxy$$

$$f_3 = wy'z + w'xyz'$$

✓ Answer ✓

1s	Binary	Function
2	0011.	1, 2
2	1001.	1, 2, 3
2	0110	3
3	0111.	1, 2
3	1011.	1, 2
3	1101.	1, 2, 3
3	1110.	2
4	1111.	1, 2

1s	Binary	Function
2	0_11.	1, 2
2	_011.	1, 2
2	10_1.	1, 2
2	1_01	1, 2, 3
2	0110	3
3	_111.	1, 2
3	1_11.	1, 2
3	11_1.	1, 2
3	111_	2

1s	Binary	Function
2	__11	1, 2
2	1__1	1, 2
2	1_01	1, 2, 3

1s	Binary	Function
2	0110	3
3	111_	2

	Functions	yz __11	wz 1__1	$wy'z$ 1_01	$w'xyz'$ 0110	wxy 111_
Functions		1, 2	1, 2	1, 2, 3	3	2
0011	1, 2	E				
1001	1, 2, 3		e	E		
0110	3				E	
0111	1, 2	E				
1011	1, 2	e	e			
1101	1, 2, 3		e	E		
1110	2					E
1111	1, 2	e	e			e

$$p_1 = yz$$

$$p_2 = wy'z$$

$$p_3 = w'xyz'$$

$$p_4 = wxy$$

$$f_1 = p_1 + p_2$$

$$f_2 = p_1 + p_2 + p_4$$

$$f_3 = p_2 + p_3$$

b

$$f_1 = wx' + x'z + xy'z'$$

$$f_2 = wz + xz' + xy + w'x'z$$

✓ Answer

1s	Binary	Function
0	0000.	1
1	0010.	1
1	0100.	1, 2
1	1000.	1
2	0110.	2
2	1001.	1, 2
2	1010.	1
2	1100.	1, 2

1s	Binary	Function
3	1011.	1, 2
3	1101.	2
3	1110.	2
4	1111.	2

1s	Binary	Function
0	00_0.	1
0	0_00.	1
0	_000.	1
1	_010.	1
1	01_0.	2
1	_100.	1, 2
1	100_.	1
1	10_0.	1
1	1_00.	1
2	_110.	2
2	10_1.	1, 2
2	1_01.	2
2	101_.	1
2	110_.	2
2	11_0.	2
3	1_11.	2
3	11_1.	2
3	111_.	2

1s	Binary	Function
0	_0_0	1
0	__00	1
1	_1_0	2
1	10__	1
1	_100	1, 2
2	1__1	2
2	11__	2
2	10_1	1, 2

	Functions	$x'z'$ _0_0	$y'z'$ __00	wx' 10__	$xy'z'$ _100	$wx'z$ 10_1	xz' _1_0	wz 1__1	wx 11__
Functions		1	1	1	1, 2	1, 2	2	2	2
0000	1	e	e						
0010	1	E							
1000	1	e	e	e					
1010	1	e		e					
0100	1, 2		O		x		e		
1100	1, 2		O		x		e		e
1001	1, 2			O		x		o	
1011	1, 2			O		x		o	
0110	2						E		
1101	2							O	o
1110	2						e		e
1111	2							O	o

$$p_1 = x'z'$$

$$p_2 = y'z'$$

$$p_3 = wx'$$

$$p_4 = xz'$$

$$p_5 = wz$$

$$f_1 = p_1 + p_2 + p_3$$

$$f_2 = p_4 + p_5$$

C

$$f_1 = y'z + w'z$$

$$f_2 = w'xy' + wy'z + wx'y + x'yz'$$

$$f_3 = xyz' + wx'z$$

✓ Answer

1s	Binary	Function
1	0001.	1
1	0100.	2
1	0010.	2
2	0101.	1, 2
2	1001.123	1, 2, 3
2	0011.	1
2	1010.	2

1s	Binary	Function
2	0110.	3
3	1101.	1, 2
3	0111.	1
3	1011.	2, 3
3	1110.	3

1s	Binary	Function
1	0_01.	1
1	_001.	1
1	00_1	1
1	010_	2
1	_010	2
2	_101.1	1, 2
2	1_01.1	1, 2
2	10_1	2, 3
2	1001	1, 2, 3
2	0_11.	1
2	101_	2
2	_110	3

1s	Binary	Function
1	__01	1
1	0__1	1
1	010_	2
1	_010	2
2	_101	1, 2
2	1_01	1, 2
2	10_1	2, 3
2	1001	1, 2, 3
2	101_	2
2	_110	3

	Functions	$wx'y'z$ 1001	$y'z$ __01	$w'z$ 0__1	$xy'z$ _101	$wy'z$ 1_01	$w'xy'$ 010_	$x'yz'$ _010	$wx'y$ 101_	$wx'z$ 10_1
Functions		1, 2, 3	1	1	1, 2	1, 2	2	2	2	2, 3
1001	1, 2, 3	x	0			x				0

	Functions	$wx'y'z$ 1001	$y'z$ __01	$w'z$ 0__1	$xy'z$ _101	$wy'z$ 1__01	$w'xy'$ 010__	$x'yz'$ __010	$wx'y$ 101__	$wx'z$ 10__1
0001	1		e	e						
0011	1			E						
0111	1			E						
0101	1, 2		O	x	O		x			
1101	1, 2		O		O	x				
0100	2						E			
0010	2							E		
1010	2							e	e	
1011	2, 3								x	O
0110	3									
1110	3									

$$p_1 = y'z$$

$$p_2 = w'z$$

$$p_3 = xy'z$$

$$p_4 = w'xy'$$

$$p_5 = x'yz'$$

$$p_6 = wx'z$$

$$p_7 = wyz'$$

$$f_1 = p_1 + p_2$$

$$f_2 = p_3 + p_4 + p_5 + p_6$$

$$f_3 = p_6 + p_7$$

2

Design a minimal-gate multiple-output two-stage network (using AND gates. OR gates) for the following functions.

$$f_1(w, x, y, z) = \Sigma(1, 4, 5, 7, 13) + d(3, 6)$$

$$f_2(w, x, y, z) = \Sigma(3, 5, 7) + d(6)$$

$$f_3(w, x, y, z) = \Sigma(3, 4, 11, 13, 15) + d(9, 14)$$

✓ Answer

$wx \backslash yz$	00	01	11	10
00		1	d	
01	1	1	1	d
11				1
10				

$wx \backslash yz$	00	01	11	10
00			1	
01		1	1	d
11				
10				

$wx \backslash yz$	00	01	11	10
00			1	
01	1			
11		1	1	d
10		d	1	

$$p_1 = w'z$$

$$p_2 = w'x$$

$$p_3 = xyz'$$

$$p_4 = p_1x$$

$$p_5 = p_1y$$

$$p_6 = wz$$

$$p_7 = p_2y'z'$$

$$p_8 = x'yz$$

$$f_1 = p_1 + p_2 + p_3$$

$$f_2 = p_4 + p_5$$

$$f_3 = p_6 + p_7 + p_8$$

3

Use the tabular method to determine a minimal multiple-output two-stage circuit (AND-OR) for the following functions.

$$f_1(w, x, y, z) = \Sigma(1, 5, 7, 8, 12, 13, 14, 15)$$

$$f_2(w, x, y, z) = \Sigma(0, 1, 5, 8, 14)$$

$$f_3(w, x, y, z) = \Sigma(0, 2, 3, 6, 8, 9, 10, 11, 14)$$

✓ Answer

1s	Binary	Function
0	0000.	2, 3
1	0001.	1, 2
1	0010.	3
1	1000.123	1, 2, 3
2	0101.	1, 2
2	1100.	1

1s	Binary	Function
2	0011.	3
2	0110.	3
2	1001.	3
2	1010.	3
3	0111.	1
3	1101.	1
3	1011.	3
3	1110.123	1, 2, 3
4	1111	1

1s	Binary	Function
0	_000.3	2, 3
0	00_0.	3
0	000_	2
1	_010.	3
1	0_01	1, 2
1	0_10.	3
1	1_00	1
1	10_0.	3
1	001_.	3
1	100_.	3
1	1000	1, 2, 3
2	_101	1
2	_011.	3
2	_110.	3
2	1_10.	3
2	01_1	1
2	10_1.	3
2	11_0	1
2	110_.	1
2	101_.	3
3	111_.	1
3	1110	1, 2, 3

1s	Binary	Function
0	0 0	3

1s	Binary	Function
0	_000	2, 3
0	000_	2
1	_01_	3
1	__10	3
1	0_01	1, 2
1	1_00	1
1	10__	3
1	1000	1, 2, 3
2	_101	1
2	01_1	1
2	11__	1
3	1110	1, 2, 3

	Functions	$wx'y'z'$ 1000	$wxyz'$ 1110	$wy'z'$ 1_00	$w'xz$ 01_1	$xy'z$ _101	wx 11__	$w'y'z$ 0_01	$w'x'y'$ 000_	$x'y'z'$ _000
Functions		1, 2, 3	1, 2, 3	1	1	1	1	1, 2	2	2, 3
1000	1, 2, 3	Eoe		eo						eo
1110	1, 2, 3		eE				eee			
1100	1			e			e			
0111	1				E					
1101	1					e	e			
1111	1						E			
0001	1, 2							o	o	
0101	1, 2				ex	eu		OO		
0000	2, 3								u	O
0010	3									
0011	3									
0110	3									
1001	3									
1010	3									
1011	3									

$$f_1 = wx'y'z' + w'xz + wx + w'x'y'$$

$$f_2 = wxyz' + w'y'z + x'y'z'$$

$$f_3 = x'y'z' + x'y + yz' + wx'$$

4

Below is a list of prime implicants for $F(a, b, c, d)$. $F(a, b, c, d)$ is completely specified (no don't care).

$$A = 1, 3, 9, 11$$

$$B = 3, 7, 11, 15$$

$$C = 9, 11, 13, 15$$

$$D = 12, 13$$

$$E = 4, 12$$

$$F = 4, 6$$

$$G = 6, 7$$

a

Compute the number of literals for prime implicant A , B , and E .

✓ **Answer**

$$A = b'd \text{ (2)}$$

$$B = cd \text{ (2)}$$

$$E = bc'd' \text{ (3)}$$

b

Complete the prime implicant table.

✓ **Answer**

	A	B	C	D	E	F	G
1	E						
3	e	e					
11	e	e	e				
9	e		e				
7		O					o
15		o	o				
13			o	O			
12				o	o		
4					o	O	
6						o	o

c

Find a minimal sum for F . You may use both column and row elimination.

✓ **Answer**

$$F = A + B + D + F$$

5

Using tagged tabulation procedure:

$$f(w, x, y) = \Sigma(0, 2, 5, 6, 7)$$

$$g(w, x, y) = \Sigma(2, 3, 5, 6, 7)$$

$$h(w, x, y) = \Sigma(0, 2, 3, 4, 5)$$

a

Find all of the multiple output prime implicants.

✓ Answer

1s	Binary	Function
0	0000.	1, 3
1	0010.123	1, 2, 3
1	0100.	3
2	0101.123	1, 2, 3
2	0110.	1, 2
2	0011.2	2, 3
3	0111.	1, 2

1s	Binary	Function
0	0_00	3
0	00_0	1, 3
1	0010	1, 2, 3
1	0_10	1, 2
1	001_	2, 3
1	010_	3
2	0101	1, 2, 3
2	0_11	2
2	01_1	1, 2
2	011_	1, 2

1s	Binary	Function
0	0_00	3
0	00_0	1, 3
1	0010	1, 2, 3

1s	Binary	Function
1	0_10	1, 2
1	001_	2, 3
1	010_	3
1	0_1_	2
2	0101	1, 2, 3
2	0_11	2
2	01_1	1, 2
2	011_	1, 2

	Functions	0010	0101	0_10	01_1	011_	0_1_	0_11	00_0	001_	0
Functions		1, 2, 3	1, 2, 3	1, 2	1, 2	1, 2	2	2	1, 3	2, 3	3
0010	1, 2, 3	o		o			o		e	e	
0101	1, 2, 3		o		o						
0111	1, 2				O	o	o	o			
0110	1, 2			O		o	o				
0000	1, 3								E		u
0011	2, 3						e	e		E	
0100	3										u

$$w'x'yz' + w'xy'z + w'yz' + w'xz + w'xy + w'x'z' + w'x'y$$

b

Compute the compact MOPI?

✓ **Answer**

$$w'yz' + w'xz + w'x'z' + w'x'y$$

c

List all essential MOPI?

✓ **Answer**

$$w'x'z' + w'x'y$$